Assignment 2: Data Preprocessing

Resource-Efficient and Explainable AI for Photovoltaic Anomaly Detection Under Sparse Sensing Constraints

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1. Scope and dataset

The current milestone covers acquisition, audit, train-only preprocessing and exploratory diagnostics. The immutable source export contains 1,373,798 observations and seven variables: vdc1/vdc2 (V), idc1/idc2 (A), irradiance irr (W/m²), module temperature pvt (°C), and nominal f_nv. Labels are 0 normal, 1 short circuit, 2 resistive degradation, 3 open circuit, and 4 natural building shade. Six predictors are continuous float64; the target is uint8 storage representing nominal categories. No categorical predictors require encoding. Downstream predictive outcomes are outside this report.

2. Source and population discrepancy

The original-author dataset is pinned at e9ed8ad8072dcb3ffc643b0f3604219798708551. Lazzaretti et al. (Sensors 20, 4688, 2020; DOI 10.3390/s20174688) provides experiment context. The paired source vectors are provisionally aligned by exported order, not verified timestamps. Filtering irr >= 100 W/m² yields 515,958 observations and literature class counts [309253, 5999, 10371, 6024, 184311]. This exact numerical match suggests a possible population explanation, but the source does not explicitly document that exact filtering operation. All raw positions are retained rather than removed to force agreement.

3. Missingness and suspicious values

All seven variables contain zero conventional missing cells: 0/9,616,586 (0%). All 1,373,798 rows are complete. No empirical MCAR/MAR/MNAR mechanism can be inferred from zero exported NaNs; omitted intervals remain unobservable. The 62 paired irr=0/pvt=0 positions, in runs of 1/32/29, remain suspicious observations rather than confirmed missingness. One irradiance observation is -0.0381 W/m². The existing near-zero convention clamps -1 <= irr < 0 to zero; irradiance below -1 and negative electrical measurements would become unavailable, while valid negative Celsius measurements are retained. The raw CSV is not altered.

4. Cleaning and imputation

No statistical outlier capping or row deletion is selected because unusual observations may contain genuine fault information. TRAIN IQR fences are Q1 - 1.5 IQR and Q3 + 1.5 IQR; |Z| > 3 is a distribution-qualified complementary screen. Existing distribution/skewness evidence supports median contingency imputation rather than outlier-sensitive means. Mode is unsuitable for continuous measurements; KNN adds distance/compute assumptions, and interpolation requires continuity unavailable in the export. Low-light medians can be unsuitable during generation, so contingency assumptions remain explicit. Retained-channel availability indicators preserve missingness information. At most one missing retained channel for at most two consecutive source positions is eligible under the existing contingency mask; multi-channel/sustained unavailability is flagged. All arrays retain row alignment.

5. Target distribution and statistical matching

The raw class counts are [1162931, 5999, 10371, 6024, 188473]. TRAIN counts are [844805, 1226, 5695, 1204, 153199], with normal approximately 83.966%. Labels remain nominal and are separated from predictors. No synthetic resampling is performed. Continuous predictor pairs use descriptive Pearson correlation; Spearman can provide a rank-based sensitivity analysis but is not newly claimed here. Continuous predictors versus the five-class nominal target use descriptive mutual information, not Pearson on integer target codes. The retained Phase 1 data-type/test strategy distinguishes continuous, binary, nominal and ordinal cases; no unsupported p-value claim is made for dependent observations.

6. Feature relationships and collinearity

Five TRAIN predictor pairs exceed |r| >= 0.85. VIF values are vdc1=52.168202, vdc2=56.345662, idc1=31.495610, idc2=15.780847, irr=39.292651, pvt=4.069798. Five exceed 10. Dependence warnings do not prove physical redundancy; no automatic predictor deletion follows. Scatterplots display a seed-42 uniform 20,000-position TRAIN sample with full-TRAIN descriptive OLS lines. Mutual information uses a seed-42 proportional stratified TRAIN sample of 50,000, k=3, continuous predictors and nominal target. Repeated measurements, ties and class imbalance limit causal or inferential interpretation.

7. Split and leakage prevention

The original ordered partitions are TRAIN [0,1006129), N=1,006,129; VALIDATION [1056990,1173967), N=116,977; TEST [1227337,1373798), N=146,461. Two separator intervals exclude 104,231 positions. Sixteen signal-defined blocks follow a 12/2/2 allocation, not verified days/sessions. The rule is irr < 100 W/m² for at least 3,600 source positions, not a verified hour. The source study describes nominal 1 Hz acquisition, per-second electrical RMS values, one-second environmental averages and a 16-day experiment, but no timestamps identify uninterrupted intervals. Naive random row splitting would mix strongly dependent neighboring measurements. The original validation sub-block roles remain planning metadata; no downstream decision is repeated here. Medians, scalers and PCA fit only on TRAIN; validation/test are transform-only. Labels, acquisition index and block identity are excluded from predictive inputs. Historical exposure means the reserved partition is not asserted to be pristine.

8. Pipeline implementation and dimensions

Scikit-Learn Pipeline and ColumnTransformer implement the existing quality rules, TRAIN-median fallback, optional target-free physical features, optional StandardScaler, and all retained-channel indicators. Three existing representations are demonstrated without comparing trained predictors: six_standard has 6 raw + 6 indicators = 12 mathematical inputs; electrical_native has 4 raw + 4 indicators = 8; electrical_derived has 4 raw + 5 physical derivatives + 4 indicators = 13. Physical channel counts remain 6/4/4. P1, P2 and P_total are RMS voltage/current products and proxies, with absolute voltage/current differences also available; each derived feature uses only retained parents. No adequacy or optimal sensing conclusion follows from these dimensions.

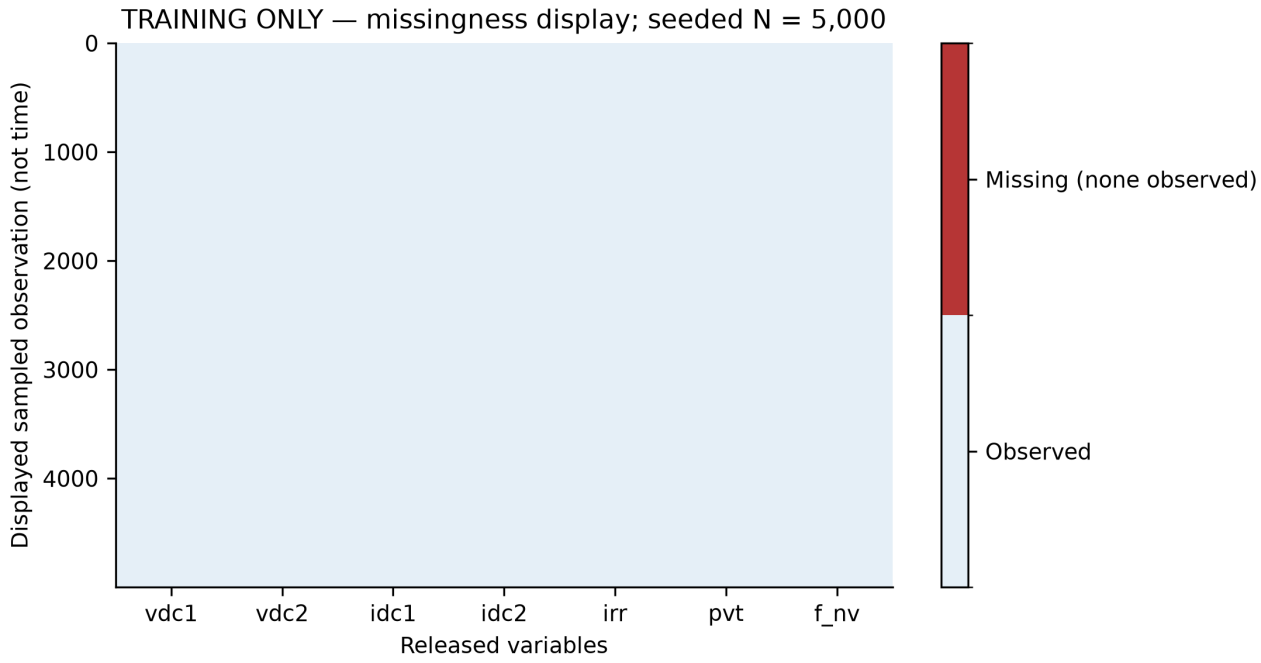
9. Scaling, transformations and PCA

StandardScaler is retained for the standardized representation because unequal units distort later regularized or distance-sensitive analysis; tree-oriented representations remain native-scale. RobustScaler and MinMaxScaler are descriptive alternatives. Scaling does not Gaussianize skewed data. No log transform is introduced. Full-TRAIN StandardScaler and full-SVD PCA explain approximately 95.1677% with the first two components; two components cross both the 90% and 95% cutoffs. PCA is mathematical dimensionality reduction, not proof that only two physical sensors are required; all six measurements are needed to compute those components.

10. Reproducibility and limitations

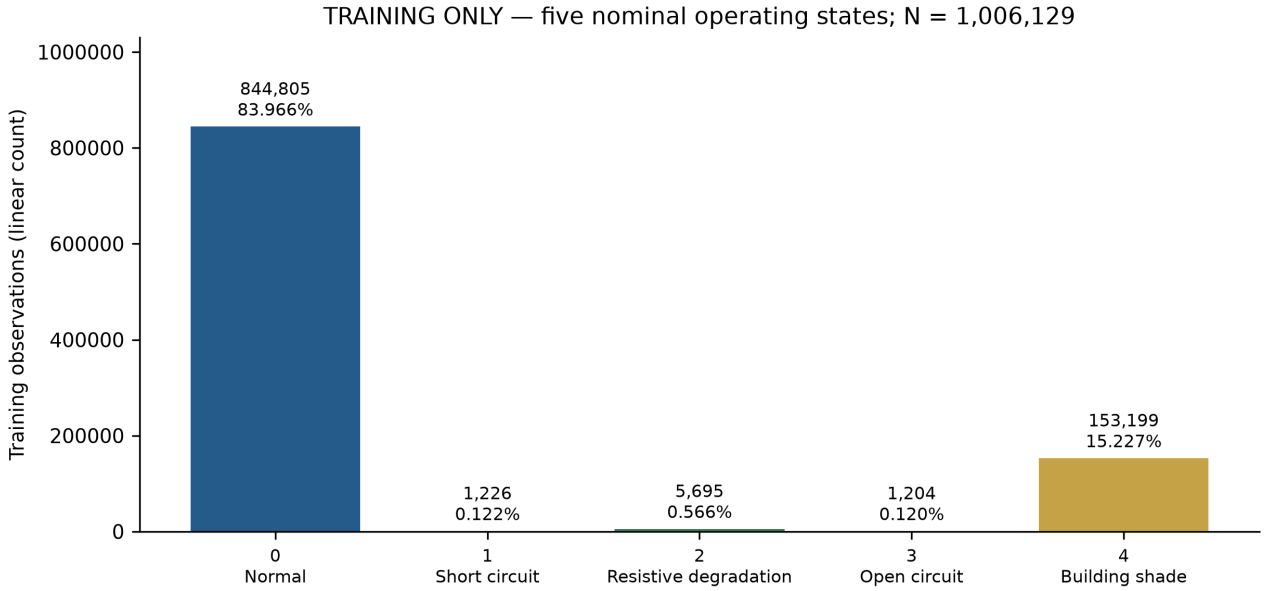
The notebook loads the immutable CSV, executes target-free preprocessing, validates output dimensions and compares descriptive statistics with the original verified reference. The eight original 300-DPI figures are retained byte-for-byte rather than regenerated. The standalone code has no model cache dependency. No new predictive experiment is included. Duplicate removal is not claimed. A single controlled Brazilian installation, dependent source positions, uncertain continuous timing, provisional paired-vector alignment and unavailable independent sites limit transferability. Further course stages may address predictive modeling; no current-stage deployment claim is made.

Diagnostic 1: Missingness pattern



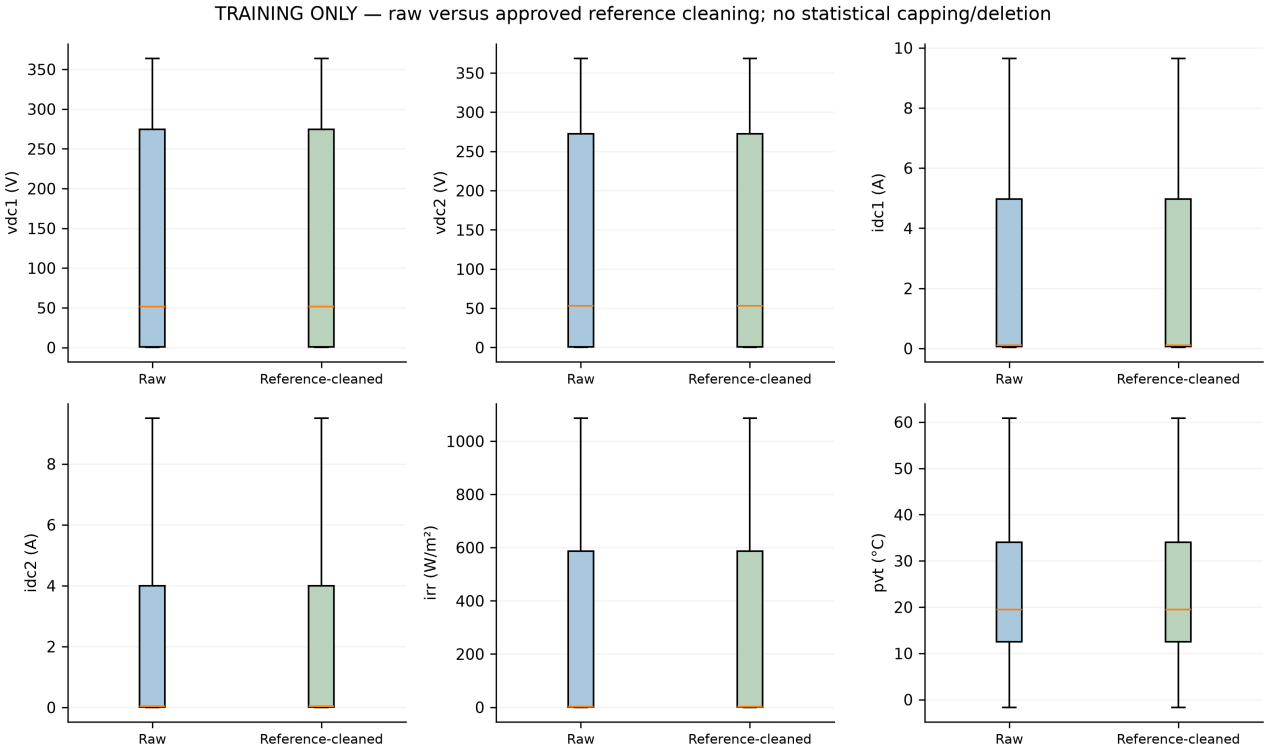
Seed-42 uniform TRAIN display. Full released-data audit: 0 conventional missing cells across all seven variables. The 62 paired zeros remain observations, not confirmed missing values.

Diagnostic 2: Target class distribution



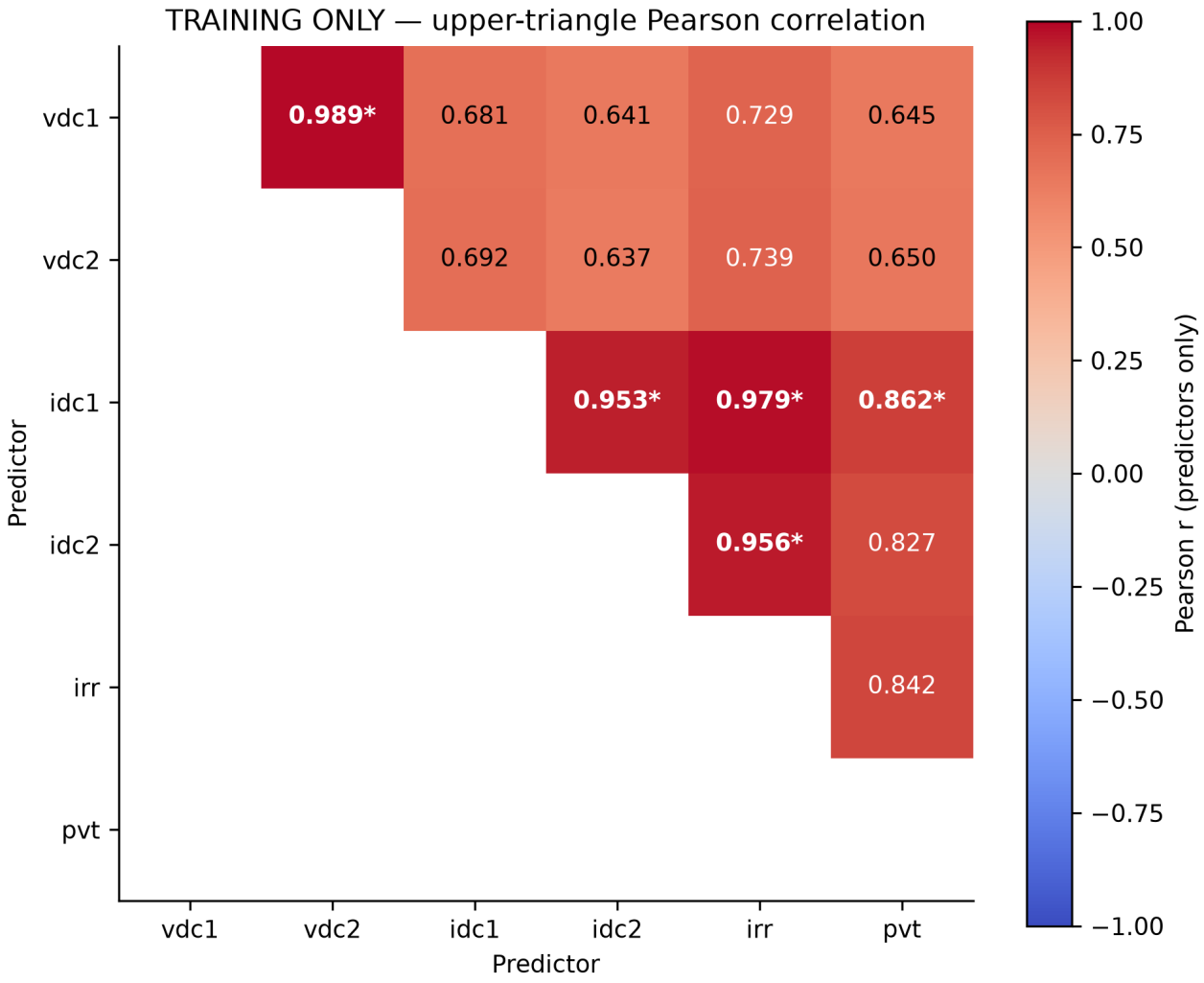
All TRAIN labels, without rebalancing. Counts and percentages expose rare hardware states; accuracy alone is inadequate. Class 4 is natural building shade, not hardware failure.

Diagnostic 3: Raw vs reference-cleaned boxplots



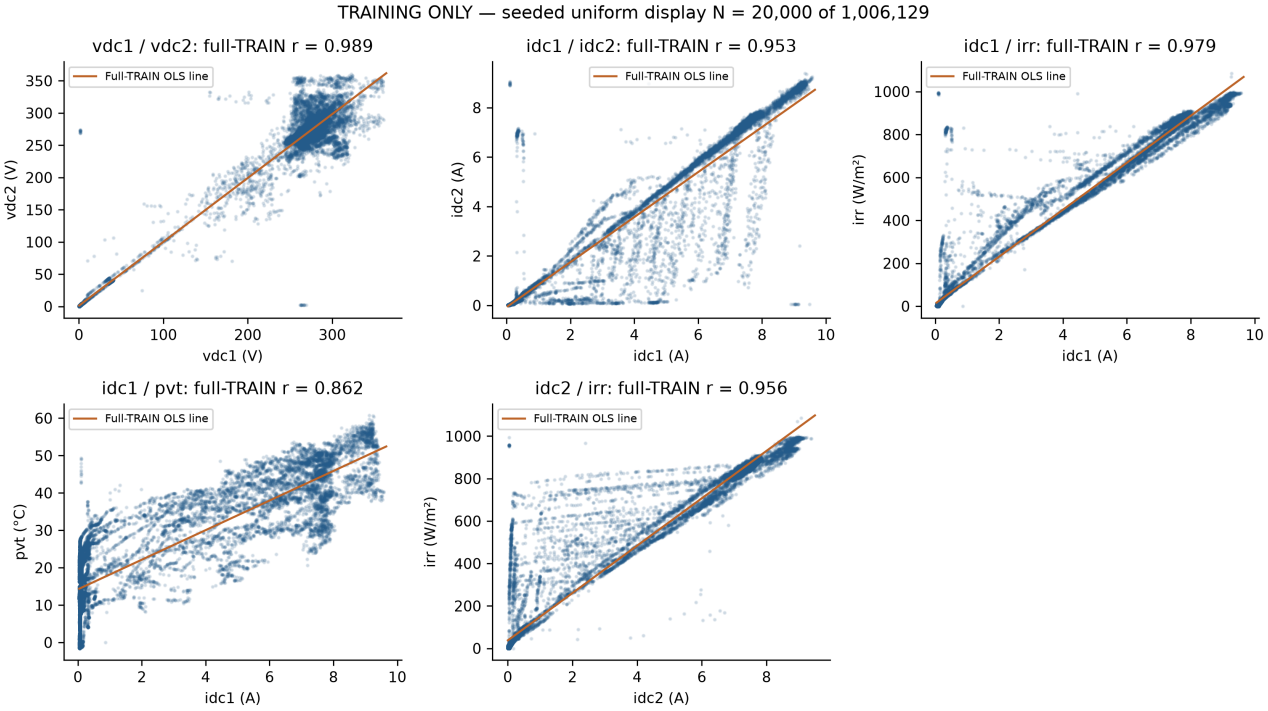
Full TRAIN IQR box statistics and fliers are shown. Reference cleaning is the existing RetainedQuality policy; unusual values remain. Nearly identical panels reflect intentional retention, not failed cleaning. Units are channel-specific.

Diagnostic 4: Training predictor correlation



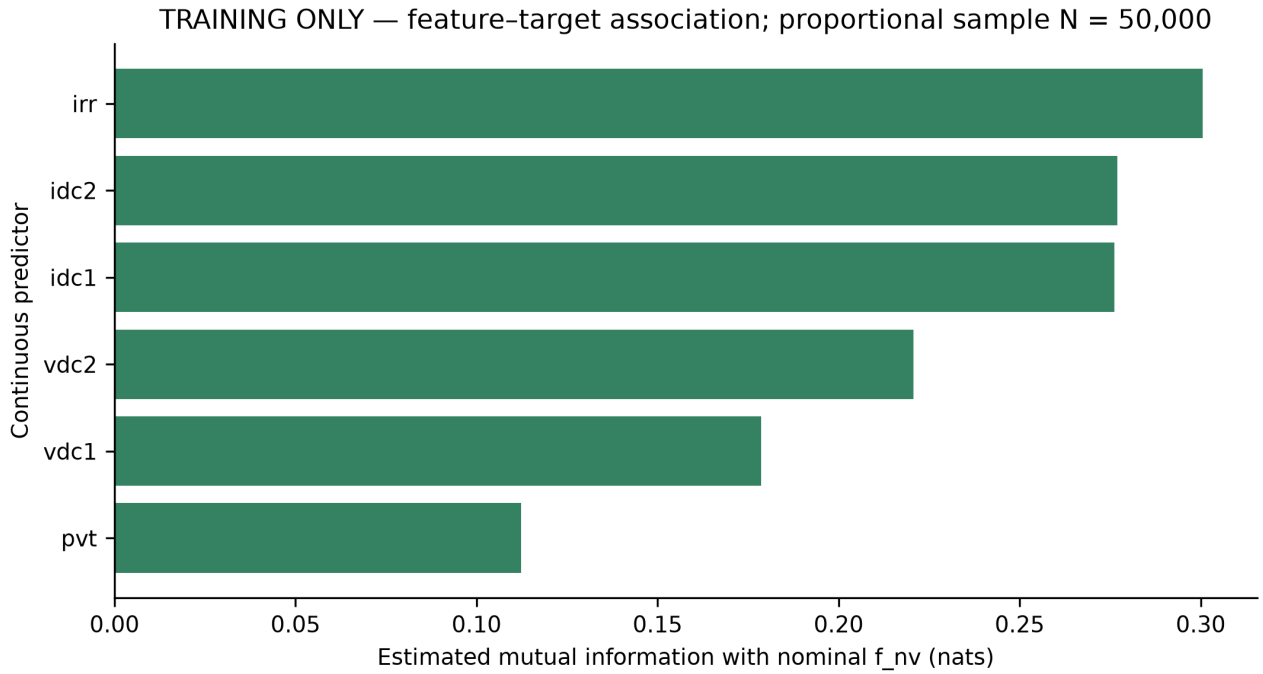
All six raw TRAIN predictors; f_nv is excluded. Asterisks mark |r| >= 0.85 (five pairs), matching Phase 1. Screening warns about redundancy; the preprocessing reference retains physical channels for possible later study, rather than automatically dropping them.

Diagnostic 5: Collinear predictor pairs



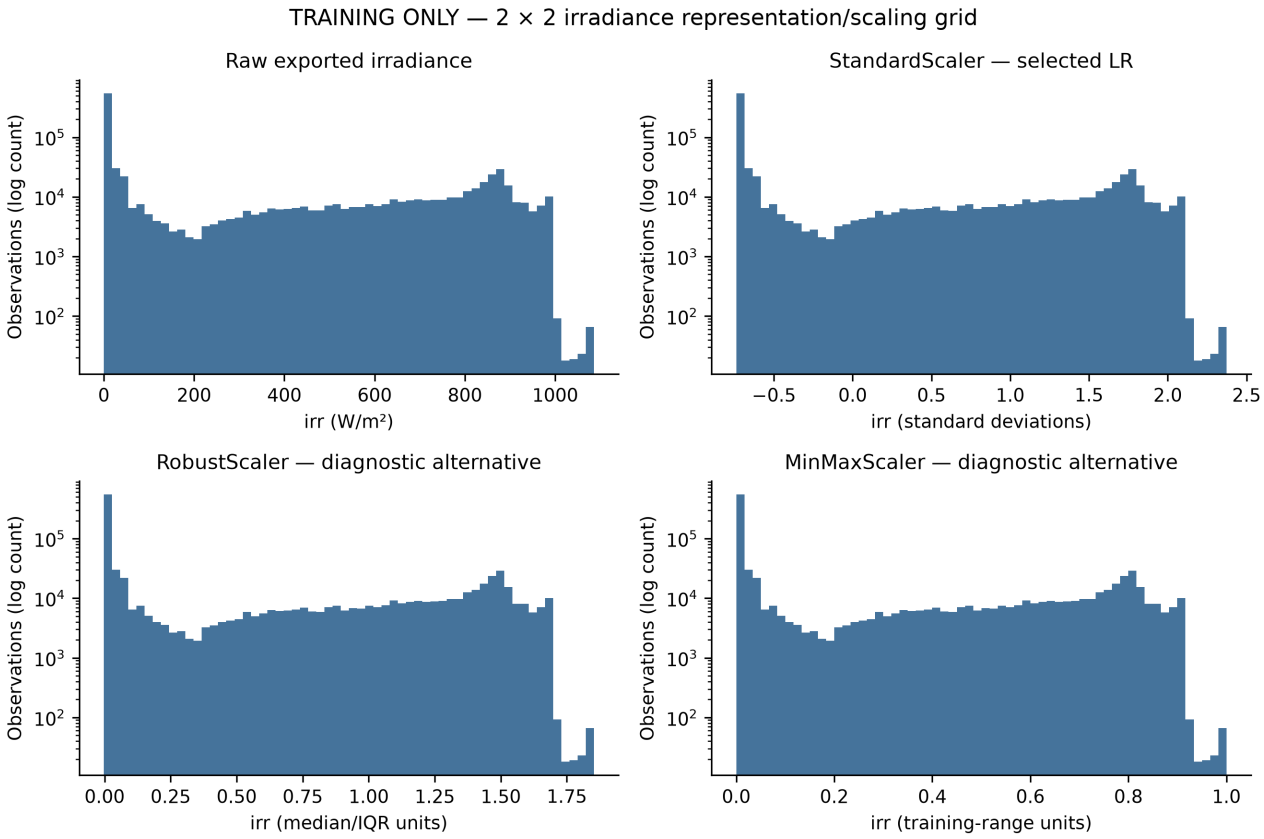
All five Phase 1 high-correlation pairs. Points are a uniform seed-42 TRAIN sample; correlations and descriptive OLS lines use full TRAIN. Association is not causation; dependence precludes naive significance claims.

Diagnostic 6: Feature-target mutual information



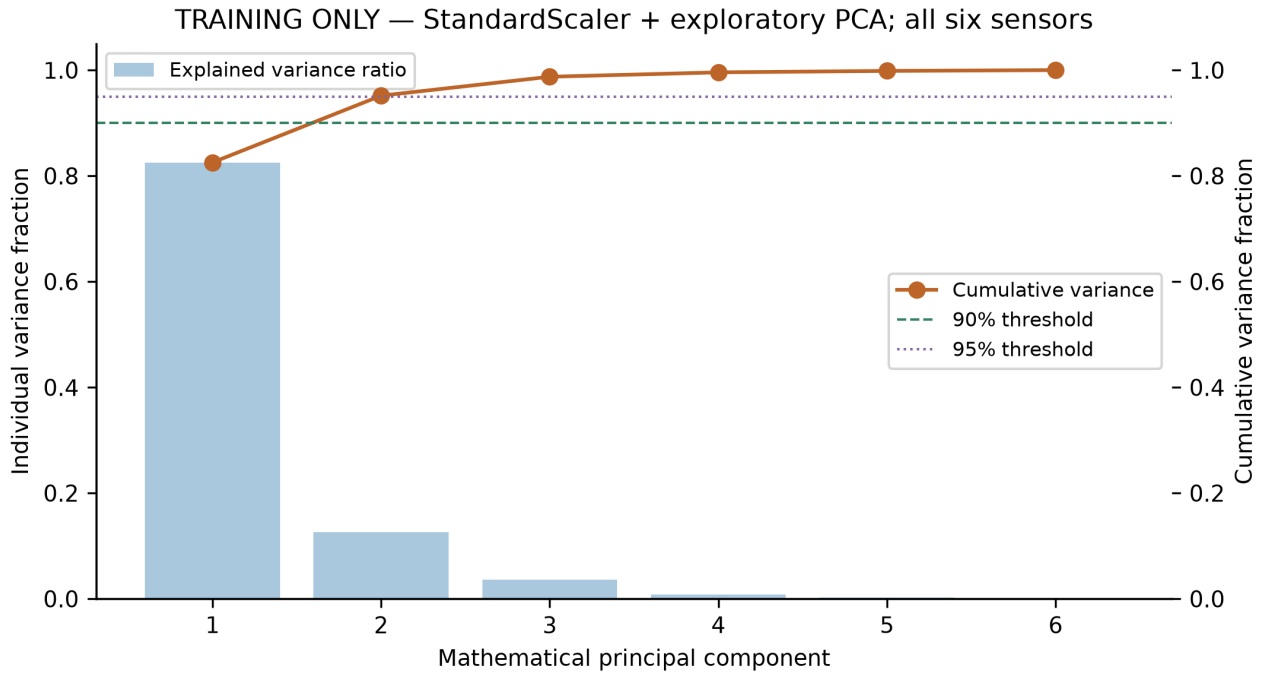
Seed-42 proportional stratified TRAIN sample, k=3 continuous-feature MI estimator; train-fitted standardization. Target codes remain nominal. This descriptive diagnostic does not alter preprocessing choices; imbalance, repeated readings and ties limit interpretation.

Diagnostic 7: Transformation/scaling comparison



All learned scaler parameters use full TRAIN. Scaling changes units, not the underlying skewed shape. StandardScaler aligns LR regularization; RF/LightGBM retain native scale. Robust/MinMax are diagnostic alternatives only. Optional log1p is not introduced into frozen pipelines.

Diagnostic 8: PCA scree and cumulative variance



StandardScaler and full-SVD PCA are fitted to full TRAIN only. PCA is an exploratory mathematical diagnostic, not physical sensor reduction, a feature-selection rule, or a change to the preprocessing policies. All six inputs are needed to compute the components.

11. Preprocessing strategy versus reality

Missing-value strategy

Phase 1 planned decision: TRAIN-median contingency plus availability indicators

Actual implementation: Same median/indicator preprocessing; exported NaNs remain zero

Status: IMPLEMENTED AS PLANNED

Deviation / finding: No missingness mechanism inferred from zero NaNs

Rationale / evidence: Preprocessing_Reference.json; src/pipeline_builder.py

Paired-zero handling

Phase 1 planned decision: Annotate suspicious simultaneous irr/pvt zeros

Actual implementation: All 62 paired-zero observations retained as observations

Status: IMPLEMENTED AS PLANNED

Deviation / finding: Not relabeled as confirmed missing values

Rationale / evidence: Phase1 primary_audit.json; compliance raw audit

Negative irradiance

Phase 1 planned decision: Near-zero convention, with physical caution

Actual implementation: Existing quality rule clamps -1 <= irr < 0; no new rule

Status: IMPLEMENTED AS PLANNED

Deviation / finding: One released slightly negative observation; convention unchanged

Rationale / evidence: src/transformers.py RetainedQuality

Negative temperature

Phase 1 planned decision: Retain physically valid negative Celsius values

Actual implementation: Negative Celsius values remain unchanged

Status: IMPLEMENTED AS PLANNED

Deviation / finding: No automatic zero-floor on module temperature

Rationale / evidence: src/transformers.py RetainedQuality

Outlier policy

Phase 1 planned decision: Screen IQR/Z values; retain plausible fault observations

Actual implementation: No statistical capping or row deletion; raw/reference boxplots

Status: IMPLEMENTED AS PLANNED

Deviation / finding: Similar before/after plots reflect intentional retention

Rationale / evidence: Training_Distributions.csv; diagnostic 3

Target encoding

Phase 1 planned decision: Treat f_nv codes as five nominal labels

Actual implementation: Nominal 0–4 labels retained; no categorical predictors encoded

Status: IMPLEMENTED AS PLANNED

Deviation / finding: No artificial ordinal or continuous target interpretation

Rationale / evidence: Variable_Inventory.csv

SMOTE decision

Phase 1 planned decision: Avoid initial row-level synthetic oversampling

Actual implementation: No synthetic oversampling or row duplication in preprocessing

Status: NOT REQUIRED

Deviation / finding: Dependent readings do not create independent fault episodes

Rationale / evidence: Phase1 strategy; current preprocessing retains observed rows

Split strategy

Phase 1 planned decision: Ordered 12/2/2 signal-block allocation with buffers

Actual implementation: Same TRAIN/VALIDATION/TEST bounds; 104231 separator rows

Status: IMPLEMENTED AS PLANNED

Deviation / finding: Source-position order; no verified day/session claim

Rationale / evidence: Phase1_Split_Manifest.json; compliance split checks

Leakage prevention

Phase 1 planned decision: Fit learned preprocessing only on TRAIN

Actual implementation: Medians/scalers fit on TRAIN; held-out arrays only transformed

Status: IMPLEMENTED AS PLANNED

Deviation / finding: New diagnostic scalers/PCA also train-only

Rationale / evidence: Preprocessing_Evidence.json fit ledger

Correlation analysis

Phase 1 planned decision: TRAIN Pearson screen |r| >= .85; no automatic deletion

Actual implementation: Same six-predictor TRAIN matrix and five flagged pairs; retained diagnostics

Status: IMPLEMENTED AS PLANNED

Deviation / finding: Correlation is not proof of physical redundancy

Rationale / evidence: Training_Correlation.csv; diagnostics 4–5

VIF

Phase 1 planned decision: TRAIN VIF >10 warning; justify physical retention

Actual implementation: Phase1 VIF audit retained: five warnings, not automatic drops

Status: IMPLEMENTED AS PLANNED

Deviation / finding: Five warnings identify dependence; no automatic predictor deletion

Rationale / evidence: primary_audit.json training_vif; Phase1 report

Mutual information

Phase 1 planned decision: TRAIN nonlinear association for nominal f_nv

Actual implementation: k=3 MI on seeded proportional TRAIN sample N=50000

Status: REFINED

Deviation / finding: Descriptive TRAIN association; not feature selection or causal evidence

Rationale / evidence: Preprocessing_Reference.json; diagnostic 6

PCA

Phase 1 planned decision: Exploratory TRAIN-scaled PCA; preserve sensor interpretation

Actual implementation: TRAIN StandardScaler + full-SVD PCA; 90%/95% thresholds

Status: REFINED

Deviation / finding: Exploratory only; not equivalent to physical sensor reduction

Rationale / evidence: Preprocessing_Reference.json; diagnostic 8

Physical feature engineering

Phase 1 planned decision: Domain-grounded derivatives with parent-channel constraints

Actual implementation: Only P1/P2/P_total/V_diff/I_diff where parents retained

Status: REFINED

Deviation / finding: Existing target-free derived-feature branch retained as preprocessing; no performance comparison

Rationale / evidence: src/transformers.py PhysicalFeatures

Scaling

Phase 1 planned decision: TRAIN-fitted standardization where scale-sensitive; native scale for trees

Actual implementation: Standardized and native-scale preprocessing branches retained; Robust/MinMax diagnostic alternatives

Status: IMPLEMENTED

Deviation / finding: No Gaussianization claim; no new log transform selected

Rationale / evidence: Preprocessing_Evidence.json; src/pipeline_builder.py

Output dimensions

Phase 1 planned decision: Record mathematical dimensions separately from physical inputs

Actual implementation: six_standard: 12; electrical_native: 8; electrical_derived: 13

Status: IMPLEMENTED

Deviation / finding: Six/four/four physical channels plus indicators and, for the derived branch, five electrical derivatives

Rationale / evidence: Preprocessing_Evidence.json; src/pipeline_builder.py

References

Lazzaretti et al., A Monitoring System for Online Fault Detection and Classification in Photovoltaic Plants, Sensors 20(17), 4688 (2020), https://doi.org/10.3390/s20174688.

Original-author dataset: https://github.com/clayton-h-costa/pv_fault_dataset, pinned commit e9ed8ad8072dcb3ffc643b0f3604219798708551; original README and Apache-2.0 license retained.

Approved Phase 1 strategy report and original audit CSV/JSON files in phase1_strategy/.

Scikit-Learn Pipeline, ColumnTransformer, preprocessing, mutual_info_classif and PCA APIs; exact existing versions in the preprocessing requirements.